TEXORA · LECTURE SHEET

Garment Printing

Screen printing and its paste families — plastisol, pigment, discharge and special effects — plus placement and all-over methods, mesh selection, curing, and the production arithmetic of a 12,000-piece print order.

AM-4101 · ADVANCED APPAREL MANUFACTURING-I · TOPIC 3 · SPECIALIZATION — APPAREL MANUFACTURING

1. Printing on garments

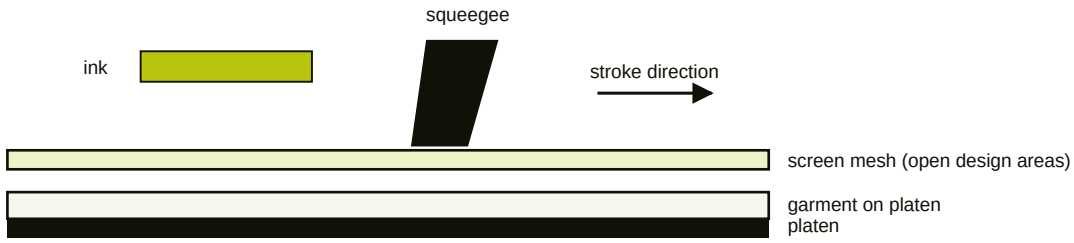
Garment printing applies colour to a defined area of a finished or part-finished garment — a chest print, a back graphic, a sleeve logo — in contrast to fabric printing, which patterns the whole cloth before cutting. Because the garment already exists, the defining operations are **placement** (registering the design at the specified position on a three-dimensional garment), **transfer through a screen or plate**, and **curing** to fix the print. Screen printing dominates garment work; digital direct-to-garment (DTG) serves short runs and photographic art; heat-transfer prints and vinyl serve names, numbers and very small lots.

2. Print pastes and their characters

SYSTEM	HOW IT FIXES	CHARACTER AND USE
Plastisol	PVC particles fused by heat (cure about 160°C)	Opaque on dark garments, sharp edges, sits on the surface; the sportswear standard
Water-based pigment	Binder film cured on the surface	Soft hand, matte, best on light grounds; less opacity on dark fabric
Discharge	Reduces the ground dye away and (in discharge-print form) deposits colour in its place	Zero-hand print on dyed cotton; shade depends on the ground's dischargeability
Rubber / high-density	Elastic binder, printed in layers	Stretchable prints; high-density builds a raised, moulded effect
Puff	Blowing agent expands during curing	Raised, spongy 3-D effect

Flock	Short fibres electrostatically planted on adhesive print	Velvet-like surface
Foil / glitter	Adhesive print + foil pressed on, or glitter particles in binder	Metallic and sparkle effects; wash durability is the limit

3. The screen-printing process



The squeegee floods, then prints: it presses ink through the open mesh onto the garment in one stroke

Figure 1 — Flat-screen printing cross-section. One screen per colour; the garment travels between screens on a manual table or a rotating automatic (octopus) machine.

The sequence per colour is: expose the design on a coated screen → mount and register the screen → load the garment on the platen at the placement marks → flood and print stroke → flash-dry if another colour follows → final cure. **Mesh selection** balances detail against ink deposit (typical planning values, threads per cm): coarse effects and glitter about 21; general plastisol 43; fine line work 77; halftone process work 90–120. A white **underbase** screen precedes colours on dark garments, so a four-colour design on black is a five-screen job.

4. Production calculation

WORKED EXAMPLE — PRINTING THE COURSE ORDER

Design: chest print, 4 colours + white underbase on dark shirts = **5 screens**. Paste consumption averages **12 g per print** for this design size.

Paste for 12,000 garments = 12 × 12,000 = 144,000 g = **144 kg**, divided among the five colours by their design coverage (the underbase usually takes the largest share).

Production rate, automatic machine ≈ 600 pcs/h → 12,000 ÷ 600 = **20 h**. Manual table ≈ 80 pcs/h → 12,000 ÷ 80 = **150 h**. The same manual-versus-automatic logic as AM-3207 Topic 1 — print-room capacity is a cycle-time calculation, and at this order size the automatic machine is the only sane plan.

5. Quality and faults

Print quality is judged on registration (colours aligned to the artwork), placement (position against the specification's measurements from neck point and centre), opacity, hand feel and durability — wash and crock (rub) fastness of the cured print, and crack testing on stretch. Characteristic faults: misregistration from a loose screen or shifted platen adhesive; pinholes from dust in the mesh; bleeding where a wet colour meets another without flash drying; and under-cured prints that wash out — the cure is verified by a wash test on first production pieces, not by the oven dial. Prints spanning seams or zips need artwork and placement designed for them from the start.

Glossary

TERM	MEANING
Platen	The flat board on which the garment is loaded for printing; its adhesive holds registration.
Underbase	A white print beneath colours on dark garments, giving them a bright ground.
Flash dry	Brief heat between colours that gels a print enough to receive the next colour.
Curing	Final heating that fully fuses/fixes the print system (plastisol about 160°C).
Mesh count	Threads per unit length of screen fabric; higher count, finer detail, thinner deposit.

Key points and summary

- Garment printing = placement + transfer + cure; screen printing dominates, with paste systems chosen by ground colour, hand feel and effect.
- Plastisol is opaque and surface-sitting; discharge removes the ground dye for a zero-hand print; special systems (puff, flock, foil) trade durability for effect.
- Screens = colours + underbase on dark garments; mesh count trades detail against deposit.
- Worked order: 5 screens, 144 kg of paste, 20 h automatic versus 150 h manual production.

Practice

Answer in your own words; brief model answers follow.

1. A design has 3 colours and is printed on white garments, then re-ordered on navy garments. How many screens does each version need, and why?

2. Paste consumption is 9 g per print for an order of 8,500 garments. Calculate total paste, and the production time on a 500 pcs/h automatic machine.
3. Why does halftone work demand a finer mesh than a glitter print?
4. A print washes out in the buyer's wash test although the curing oven showed the correct temperature. Give a plausible cause.

Answer key

1. On white: 3 screens (one per colour). On navy: 4 — dark grounds need a white underbase beneath the colours.
2. Paste = 9 × 8,500 = 76.5 kg. Time = 8,500 ÷ 500 = 17 h.
3. Halftone detail lives in tiny dots that a coarse mesh cannot reproduce (the threads block and distort them), while glitter particles are physically large and cannot pass a fine mesh at all — mesh must fit both the image and the ink.
4. The dial shows oven air temperature, not print temperature: a fast belt, a thick garment area or a cold start can leave the print film itself below fusion temperature. Cure must be verified on the print, by wash/stretch testing.

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